

CORPORATE DATA CLEANING & ETL PIPELINE REPORT

Power Query Documentation for Customers, Transactions, and Stock Datasets

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Project Type: Data Transformation & Integration	Framework: Relational Database Design

1. Executive Summary

This case study documents an end-to-end Extract, Transform, Load (ETL) pipeline engineered to structurally clean, standardize, and reconcile three core corporate tracking modules: Customer Records, Sales Transaction Ledgers, and Historical Stock Market Volume Logs. All data operations were executed systematically using the Excel Power Query data transformation engine to ensure an automated, repeatable data cleaning process.

2. System Pipeline Architecture

The table below details the specific data quality challenges encountered within each individual dataset module and the corresponding automated workflows built within Power Query to resolve them before consolidating into a final relational workbook model.

Dataset Target	Identified Quality Anomalies	Power Query Resolution
1. Customer Records	- Extra blank header spacing rows. - Misaligned database column parameters. - Structural null values within geographic regions.	- Top row removal filtering. - Dynamic column header promotion. - Implemented "Unknown" flags for text data integrity.
2. Sales Transactions	- Split ledger files across multiple database tables. - Critical missing numerical values in transaction rows. - System-serial formatted date timestamps.	- High-integrity Query Appending. - Automated Statistical Median Imputation. - Proper chronological Date type conversions.
3. Stock Logs	- Corrupt/placeholder numeric volumes ('99999999'). - Duplicate tracking instances for identical trading days.	- Conditional value overrides. - Distinct row constraints based on date keys.

Strategic Highlight: Statistical Median Imputation

When confronting null parameters in numeric arrays, discarding records entirely or substituting a basic mean can warp descriptive metrics. This pipeline calculated and applied a column-level **Median Imputation** to seamlessly patch empty transaction amounts, preserving true structural variance without skewing totals.

3. Customer Directory Data Pipeline

This page documents the successfully transformed Customer table within the Power Query interface. Structural gaps, extra space padding rows, and misaligned headings were systematically removed, and text columns were forced into uniform system properties.

The screenshot displays the Power Query interface. The main area shows a table with 20 rows of customer data. The columns are: CustomerID, CustomerName, JoinDate, Region, and SalesRep. The data is as follows:

	CustomerID	CustomerName	JoinDate	Region	SalesRep
1	CUST0001	Customer 1	16-08-2020	Asia	Rep 4
2	CUST0002	Customer 2	21-02-2020	Europe	Rep 5
3	CUST0003	Customer 3	17-07-2021	North America	Rep 1
4	CUST0004	Customer 4	16-05-2021	unknown	Rep 4
5	CUST0005	Customer 5	04-02-2021	Asia	Rep 1
6	CUST0006	Customer 6	10-12-2020	Europe	Rep 5
7	CUST0007	Customer 7	28-07-2020	Europe	Rep 3
8	CUST0008	Customer 8	27-06-2020	Europe	Rep 5
9	CUST0009	Customer 9	14-05-2022	North America	Rep 3
10	CUST0010	Customer 10	03-06-2020	North America	Rep 5
11	CUST0011	Customer 11	03-02-2020	Europe	Rep 2
12	CUST0012	Customer 12	07-10-2020	North America	Rep 1
13	CUST0013	Customer 13	23-03-2021	North America	Rep 1
14	CUST0014	Customer 14	21-04-2021	Europe	Rep 2
15	CUST0015	Customer 15	31-10-2022	North America	Rep 3
16	CUST0016	Customer 16	24-02-2020	Europe	Rep 1
17	CUST0017	Customer 17	02-11-2021	Europe	Rep 2
18	CUST0018	Customer 18	05-09-2022	Asia	Rep 1
19	CUST0019	Customer 19	27-03-2021	Europe	Rep 4
20	CUST0020	Customer 20	07-08-2022	North America	Rep 3

The 'Query Settings' pane on the right shows the following details:

- PROPERTIES**
 - Name: Table1
 - All Properties
- APPLIED STEPS**
 - Source
 - Changed Type
 - Filtered Rows
 - Promoted Headers
 - Changed Type1
 - Replaced Value
 - Removed Duplicates
 - Changed Type2** (highlighted)

4. Master Transactions Ledger Pipeline

This section displays the master appended transaction ledger. Individual data tables (``Table2`` and ``Transaction_2``) were vertically appended to create a continuous chronological ledger. It demonstrates converted calendar date configurations and the median-imputed value patch at row 17.

Queries [3]

- Table1
- Transaction_2
- Append1**

Query Formula: = Table.Distinct(#"Changed Type", {*"TransactionID"})

	A ^B C TransactionID	A ^B C CustomerID	TransactionDate	1.2 Amount	A ^B C Currency	A ^B C Product
1	TXN00001	CUST0015	05-08-2022	5526.59	EUR	Product C
2	TXN00002	CUST0006	19-11-2022	9725.24	EUR	Product C
3	TXN00003	CUST0012	06-08-2022	8677.41	USD	Product B
4	TXN00004	CUST0006	27-05-2023	609.07	USD	Product B
5	TXN00005	CUST0012	10-02-2023	7346.86	USD	Product C
6	TXN00006	CUST0012	15-04-2023	6976.25	JPY	Product B
7	TXN00007	CUST0007	27-05-2022	5601.22	USD	Product A
8	TXN00008	CUST0009	29-09-2022	3034.84	USD	Product A
9	TXN00009	CUST0006	23-05-2022	6589.14	JPY	Product A
10	TXN00010	CUST0003	10-09-2022	1559.75	EUR	Product A
11	TXN00011	CUST0020	29-07-2023	4630.27	USD	Product B
12	TXN00012	CUST0006	06-07-2023	4810.38	JPY	Product A
13	TXN00013	CUST0006	27-09-2022	9561.25	USD	Product C
14	TXN00014	CUST0018	15-03-2023	8820.6	USD	Product C
15	TXN00015	CUST0008	13-02-2023	3002.2	JPY	Product A
16	TXN00016	CUST0006	06-01-2023	5601.22	EUR	Product C
17	TXN00017	CUST0015	13-08-2022	2197.19	JPY	Product A
18	TXN00018	CUST0013	22-05-2022	9169.96	USD	Product A
19	TXN00019	CUST0009	06-06-2023	8769.93	EUR	Product A
20	TXN00020	CUST0006	21-05-2023	3335.23	JPY	Product C
21	TXN00021	CUST0018	04-04-2022	6570.02	JPY	Product C
22	TXN00022	CUST0008	18-02-2022	6285.22	EUR	Product A
23	TXN00023	CUST0011	23-04-2022	1951.47	USD	Product B

Query Settings

PROPERTIES

Name

All Properties

APPLIED STEPS

- Source ✖
- Replaced Value ✖
- Uppercased Text
- Changed Type
- X Removed Duplicates**

5. Historical Stock Volume Log Pipeline

This page documents the cleaning sequence of the historical stock market log. The view captures the data state following the removal of duplicate trading dates and the programmatic replacement of the extreme volume placeholder outlier (`99999999`).

Queries [1]
Table1

Formula Bar: `= Table.Distinct("#Replaced Value", {"Date"})`

	Date	1.2 Open	1.2 High	1.2 Low	1.2 Close	1.2 Volume
1	01-08-2024	133.45	126.55	107.43	102.53	
2	02-08-2024	110.71	126.18	104.51	149.96	
3	03-08-2024	106.62	128.35	117.43	141.8	
4	04-08-2024	146.78	141.45	116.09	148.45	
5	05-08-2024	128.55	138.67	108.93	146.32	
6	06-08-2024	123.63	154.21	107.49	142.43	
7	07-08-2024	139.23	109.92	141.16	108.32	
8	08-08-2024	140.37	125.13	117.16	124.28	
9	09-08-2024	109.52	121.97	138.07	110.69	
10	10-08-2024	104.85	148.08	122.52	120.05	

Query Settings

PROPERTIES

Name
Table1

All Properties

APPLIED STEPS

Source
Changed Type
Changed Type1
Replaced Value ✱
X Removed Duplicates